

Checkout Chaos — Game Design Doc

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Overview

Checkout Chaos is a 3D shopping-cart racer in Godot. One player and three bot shoppers race through a grocery store’s grand-opening sale, grab items, ram each other to steal hauls, and drive out the front doors to check out.

Engine	Godot 4, 3D
Camera	Third-person chase cam behind the cart
Players	1 human vs 3 bots (multiplayer is a stretch goal)
Match	Best of 3 rounds, 2 minutes each
Art	Low-poly, bright flat colors
Team	4 people, one system each
Prototype	The browser version (Three.js) we already built

Win condition: the most money checked out across 3 rounds. Items still in your cart when a round ends don’t count.

Storyline

Base premise: you’re a devoted member of the store, and your **Shopper ID card** proves it. The store is throwing a huge sale, and you plan to leave with more than anyone else. Below are six directions that build on that. Pick one, or combine them.

The Shopper ID card

The card is both a story prop and UI. It appears on the title screen, the character select, and the results screen.

- **Front:** store logo, your name, photo, member number, “Member since” year, tier badge (Bronze / Silver / Gold / Platinum)

- **Back:** barcode, lifetime savings, and a stamp row that fills with one stamp per round won
- Each bot has its own card, which doubles as its intro and personality blurb

Direction A: Grandma's Card (recommended)

Your grandma was the store's most famous shopper, a legend of past sales. She left you her Platinum Shopper ID card and one instruction: "Don't let Carl win." The bots are her old rivals, and their grandkids.

- **Why it works:** it ties the story to the inheritance theme. You inherit the card and the rivalry, and in gameplay you inherit other carts' items.
- **Round framing:** each round is one day of the sale weekend. Winning a round adds a stamp to Grandma's card.
- **Ending:** win the match and the card is reissued in your name.

Direction B: Loyalty Legend

You're one points level short of the store's mythic "Diamond" tier. The grand opening triples points, so this is your shot.

- **Round framing:** Round 1 is Bronze to Silver, Round 2 is Silver to Gold, Round 3 is the Diamond run.
- **Hook:** your card visibly upgrades between rounds. Bots are also chasing Diamond.

Direction C: Everything Must Go

The beloved neighborhood store is closing, and the final sale is 90% off. Regulars show up with their old ID cards for one last run.

- **Tone:** funny but a little bittersweet. The results screen could show "Thanks for 30 years of shopping."
- **Round framing:** the last three days before closing, with the shelves emptier each day.

Direction D: The Once-a-Decade Sale

The store holds its legendary "Doorbuster" sale once every 10 years. Shoppers train for it. You've waited your whole life.

- **Tone:** over-the-top sports movie. A training montage on the menu, an announcer on the PA, and rival shoppers with nicknames and stats on their cards.

- **Round framing:** Qualifier, Semifinal, Final.

Direction E: Expiring Card

Your membership card expires at midnight, along with 10 years of saved points. You have to spend it all before the store closes.

- **Hook:** the round timer is also your card's countdown. The ID card shows the expiration date on screen.
- **Round framing:** 3 hours before midnight, then 2, then 1.

Direction F: Shopper of the Year

The store crowns one "Shopper of the Year," whose face goes on the wall and whose card becomes a lifetime pass. Four finalists enter.

- **Hook:** a reality-show frame with a host on the PA, a confessional-style bot intro before each round, and a trophy at the end.

Store name ideas

Bumper Crop Market (the prototype name), Shop-N-Dash, Big Basket, Aisle Nine Foods, FreshWay, CartMart.

The rivals

These bot characters fit any direction.

Shopper	Card tier	Style	Personality
Coupon Carl	Gold	Rammer	Clips every coupon, and he'll run you over for a deal
Aunt Bev	Platinum	Safe banker	Has shopped here since opening day and checks out early and often
Rolling Rita	Silver	Speed demon	Boosts everywhere and holds out for a giant haul

Core loop and match structure

The core loop is grab, protect, check out, repeated as many times as you dare in 2 minutes.

```

flowchart LR
    A[Doors open] --> B[Grab items<br/>in aisles]
    B --> C{Crash?}
    C -- faster --> D[Inherit their items]
    C -- slower --> E[Lose your cart]
    D --> B
    E --> B
    B --> F[Drive out front doors]
    F --> G[Items banked]
    G --> B
  
```

Every trip back into the store is a risk-and-reward choice. A fuller cart is worth more but slower, so it's easier to catch and rob.

One round (2:00)

1. **Countdown (3 s):** carts line up outside, and the doors slide open on GO.
2. **Rush (0:00 to 1:40):** grab items, and avoid or hunt rivals. Hazards start appearing.
3. **Final call (last 20 s):** the PA announces closing, and the timer turns red. Anything not checked out will be lost.
4. **Close:** the doors shut, and the round receipt shows each shopper's banked total.

Best of 3

- Each round is scored by money banked. The round winner gets a stamp on their Shopper ID card.
- **Match winner:** the most stamps. If there's a tie, the highest total banked across all rounds wins.
- Rounds get harder: more hazards, a shorter timer (2:00, 1:45, 1:30), and bots that get more aggressive.
- Between rounds there's a 10-second results screen showing the receipt and standings. Carts reset empty for each round.

The four systems

The game is built as four systems, one per teammate. None of them works alone, and each one's output is another's input.

#	System	Owner	Owns	Depends on
1	Player	[Name]	Input, camera, player feedback	Cart (to drive), Store (round state)
2	Cart	[Name]	Movement physics, inventory, collisions, the steal rule	Player or Bot (driver input), Store (items, checkout zone)
3	Rivals	[Name]	Bot AI: targeting, pathing, ramming, banking	Cart (to drive), Store (item list, exit, timer)
4	Store / Round Manager	[Name]	Timer, doors, spawns, Deal of the Day, hazards, checkout, scoring, rounds	Cart (who holds what), all drivers (to start and stop them)

1. Player

Turns what the human presses into driving commands, and makes it feel good.

- **Controls:** gas, brake/reverse, steer, boost. The keyboard uses WASD or arrows plus Shift; gamepad support too.
- **Chase camera:** follows behind the cart, and FOV widens during boost. The camera shakes when you get rammed.
- **Feedback:** "Rammed! +7 items" and "Checked out \$180" popups, sounds, and controller rumble.
- **Output:** a `DriveCommand` (throttle, brake, steer, boost) sent to its Cart every frame.

2. Cart

The physical cart and everything it carries. Players and bots use the same Cart scene.

- **Movement:** arcade physics, with a top speed of about 15 m/s. Each item slows the cart by 1.2%, so a full cart is about 29% slower.
- **Boost meter:** +8 m/s top speed while held. It drains in about 2 s and refills in about 8 s.
- **Inventory:** up to 24 items, shown stacked in the basket, with the total value tracked.

- **Crash rule:** the steal rule (see Inheritance). The loser is stunned for 0.7 s and immune for 1.6 s so it can't be chain-robbed.
- **Signals out:** `item_collected`, `cart_robbed(from, to, items)`, `cart_full`.

3. Rivals (bots)

Three AI shoppers that drive the same Cart through the same `DriveCommand` a player uses.

- **Decisions (every 0.3 s):** pick a target by value divided by distance; chase a rival carrying a big load; or head for checkout when greedy enough or when time is short.
- **Pathing:** Godot `NavigationAgent3D` on a baked navmesh of the store.
- **Personalities:** each bot has greed (how many items before banking), aggression (how often it rams), and boost habit. Values come from the rivals table in Storyline.
- **Unsticking:** if the bot barely moves for 1 s, it reverses and turns.
- **Difficulty:** aggression rises each round.

4. Store / Round Manager

The referee and the world. It owns everything the carts compete over.

- **Round flow:** countdown, door animation, timer, final call, close, results, next round, match end.
- **Item spawns:** 6 aisles with values from \$5 to \$40 (table below). Up to 46 items on the floor, and a new one every 0.5 s.
- **Deal of the Day:** a gold \$100 item every 14 to 22 s, with a light beam and a PA announcement.
- **Hazards:** see the list below.
- **Checkout:** the zone past the front doors. It banks a cart's value and empties it.
- **Scoring:** round receipts, stamps, and match standings.

Aisle	Items	Value each	Spawn weight
1	Produce	\$5	High
2	Bakery	\$10	High
3	Dairy	\$10	High
4	Snacks	\$15	Medium

Aisle	Items	Value each	Spawn weight
5	Frozen	\$20	Low
6	Electronics	\$40	Rare
Any	Deal of the Day	\$100	1 at a time

Store hazards

Hazard	What happens	Counterplay
Wet floor spill	Steering stops working for 1 s and the cart slides	Yellow sign warns you; steer around it
Employee with pallet jack	Crosses an aisle slowly, blocks it and bumps carts	Wait or take another aisle
Falling display	A stack of cans topples, and the tile is blocked for 5 s	A wobble animation warns 1 s before
Sample lady	Slows carts that pass her (free sample!)	Go around, or accept the slowdown
Cleanup on aisle X	The PA closes an aisle for 10 s	Reroute; bots reroute too

Inheritance

Inheritance is the main way wealth moves in the game. When two carts crash, the faster cart inherits the slower cart's entire haul, and whatever doesn't fit spills onto the floor for anyone to inherit.

Rule 1: ram-steal

1. Two carts touch.
2. Compare speeds. If the faster cart is going at least 5 m/s and at least 1.5 m/s faster than the other, it wins.
3. The slower cart loses all its items, which fly across into the faster cart.
4. The slower cart is knocked back, stunned for 0.7 s, and immune for 1.6 s.
5. If the speeds are too close, both carts bounce off and nothing changes hands.

Rule 2: spilled items

- If the winner's cart can't hold everything (24-item cap), the overflow spills on the floor around the crash.
- Spilled items become normal pickups that anyone can grab, including the cart that just lost them.
- A spilled Deal of the Day stays gold and keeps its \$100 value.

Why it matters

- **Parking is dangerous.** A stopped cart has speed 0, so any moving cart can take everything.
- **Weight is a trade-off.** A full cart is slower, which makes it easier to catch and inherit from.
- **Comebacks.** The last-place shopper can take the leader's full cart with one well-timed hit.
- **The story matches.** In Direction A, you literally inherit Grandma's card and her rivals.

Worked example

	Rita (before)	You (before)	Rita (after)	You (after)
Speed	14 m/s (boosting)	6 m/s	10.5 m/s	Stunned
Items	20 (\$260)	8 (\$110)	24 (\$310)	0
Floor				4 items spill beside you

You're stunned for 0.7 s, then you can grab back the 4 spilled items before Rita turns around.

How the systems connect (Godot)

Each system is its own scene and script with a small, agreed interface, so all four people can build in parallel from day one.

flowchart LR

```

P[1. Player<br/>PlayerController] -- DriveCommand --> C[2.
Cart<br/>Cart.tscn]
B[3. Rivals<br/>BotController] -- DriveCommand --> C
S[4. Store<br/>RoundManager autoload] -- items, hazards, timer --> B
C -- item_collected, cart_robbed --> S
S -- round_started, round_ended --> P
S -- round_started, round_ended --> B

```

Player and Rivals both produce the same `DriveCommand`, and Cart doesn't know or care which one is driving. That's the key contract.

Shared data contract (agree on day one)

Name	Type	Fields	Made by	Used by
<code>DriveCommand</code>	Resource	throttle, brake, steer (-1 to 1), boost	Player, Rivals	Cart
<code>ItemData</code>	Resource	category, value, mesh, is_deal	Store	Cart, Rivals
<code>CartState</code>	Script vars	speed, items[], value, is_stunned	Cart	Store, Rivals, HUD
<code>RoundManager</code>	Autoload	time_left, round, doors_open, pickups[], scores	Store	Everyone

Signals

Signal	Emitted by	Listened to by
<code>round_started(round)</code>	Store	Player, Rivals
<code>round_ended(results)</code>	Store	Player, Rivals, UI
<code>item_collected(cart, item)</code>	Cart	Store (remove pickup), Player (sound)
<code>cart_robbed(winner, loser, items, spilled)</code>	Cart	Store (spawn spills), Player (popup), Rivals (retarget)

Signal	Emitted by	Listened to by
<code>checked_out(cart, value)</code>	Store	Player (popup), UI (scoreboard)
<code>hazard_spawned(hazard)</code>	Store	Rivals (reroute)

Working in parallel

- Each owner builds a **test scene** with fake inputs: for example, Cart driven by a scripted `DriveCommand`, or Rivals chasing dummy pickups.
- One shared `Main.tscn` is only edited at merges, to avoid conflicts.
- Merge order: Cart + Player first (drivable), then Store (round loop), then Rivals (competition).

Controls, camera, art and audio

Controls

Action	Keyboard	Gamepad
Gas	W / Up	Right trigger
Brake / reverse	S / Down	Left trigger
Steer	A D / Left Right	Left stick
Boost	Shift / Space	A (Xbox) / Cross (PS)
Pause	Esc	Start

Camera

Chase cam about 8.5 m behind and 5.5 m above the cart, looking slightly ahead. It widens from 62° to 72° field of view during boost, and it shakes briefly when you're rammed.

Art direction

- Low-poly shapes with bright flat colors and no textures except signage.

- Each aisle has its own color: green produce, orange bakery, white dairy, red snacks, blue frozen, purple electronics. Players read value by color.
- Checkered mint-and-cream linoleum floor, hanging aisle signs, and a red-and-yellow grand opening banner outside.
- Each shopper's cart has a colored rim, handle and flag, plus a floating name tag. The player is yellow.

HUD

- Timer and round number (top left), store minimap under it
- Scoreboard with banked totals and current cart values (top right)
- Cart panel: item count out of 24, cart value, boost meter (bottom center)
- Event feed of steals and checkouts (bottom left)
- Results screen styled as a store receipt, with the Shopper ID card and stamps

Audio

- Upbeat supermarket muzak that speeds up in the final 20 s
- PA announcer for doors opening, Deal of the Day, hazards, and final call
- Sound effects: item pickup blip, crash thud, steal whoosh, register ding on checkout, squeaky wheel loop

Scope

Core is what we must ship. Stretch goals only start once the core runs end to end.

Feature	Tier	System
Drivable cart with chase cam	Core	Player, Cart
24-item inventory, weight slowdown	Core	Cart
Ram-steal and spilled items	Core	Cart, Store
Boost meter	Core	Cart, Player
3 bots with personalities	Core	Rivals
6 aisles, item spawns, checkout zone	Core	Store

Feature	Tier	System
Deal of the Day	Core	Store
Best of 3 rounds, receipt results	Core	Store
Store hazards (wet floor, pallet jack, falling display)	Core	Store, Rivals
Shopper ID card screens and storyline intro	Core	Player (UI)
Sample lady, aisle closures	Stretch	Store
Cart types (heavy, fast, big basket)	Stretch	Cart
Local split-screen, 2 players + 2 bots	Stretch	Player, Store
Online multiplayer	Stretch (future)	All

Multiplayer path

The `DriveCommand` design makes multiplayer a smaller jump later. A second human is just another source of `DriveCommand`s.

1. **Local split-screen:** two `PlayerController`s with two cameras and two viewports. No networking needed.
2. **Online:** Godot's high-level multiplayer (`MultiplayerSynchronizer`), where the host runs the Store and all collisions, and clients send only their `DriveCommand`. This keeps things fair because one machine decides who was faster.

Biggest risks

- **Bot pathing through aisles.** Fallback: hand-placed waypoints at the aisle ends, like the prototype.
- **Collision feel.** Physics-based carts can jitter. Fallback: kinematic carts with our own speed rule, like the prototype.
- **Merge conflicts in shared scenes.** Fallback: only the Store owner edits `Main.tscn`.

Requirement check

Both requirements pass. The systems requirement is solid, and inheritance is clear in the mechanics, but a strict grader could call ram-steal "stealing." The story direction we pick decides how convincing it is.

Requirement	Verdict	Grade
3+ dependent systems	Pass	9 / 10
Inheritance theme	Pass, with one risk	7.5 / 10 (8.5 with Direction A)

1. Three or more dependent systems: 9 / 10

What works

- There are four systems, and each has one owner and a clear job.
- The dependencies are real. Cart does nothing without a driver. Player and Rivals have nothing to do without the Store's items. The Store can't score without the Cart's inventory.
- The shared `DriveCommand` and the signal list make the dependencies concrete, not just claimed.

What loses a point

- Player is the thinnest system: input and camera are small next to bot AI or the round manager. A grader could see it as "just controls."
- **Fix:** give Player ownership of the full HUD, the Shopper ID card screens, and game feel (popups, shake, sound). The Scope table already assigns the ID card screens to Player.

2. Inheritance: 7.5 / 10

What works

- Inheritance *is* the core mechanic, not decoration. Every crash moves an entire haul from one cart to another.
- Spilled items add a second layer: what the winner can't hold becomes an inheritance anyone can claim.
- It changes how people play: parking is risky, a heavy cart becomes a target, and comebacks are possible.

What loses points

- Inheritance normally means receiving something when someone leaves or passes it on. Taking items by ramming reads more like stealing than inheriting.
- Nothing is inherited across rounds or generations, since carts reset empty each round.

How to raise it

- **Pick Direction A (Grandma's Card).** You inherit the card, the rivalry and the legacy. That gets the grade to about 8.5.
- **Change the wording in the game and the pitch.** A rammed cart is "knocked out of the sale," and its haul is "left to" the winner. The feed can say "Rita inherited Carl's cart."
- **Optional, small change:** the round winner's Shopper ID card passes a perk to the next round (for example, +2 cart slots). That adds generational inheritance without a new system.

Open questions

- ☐ Team names: who owns Player, Cart, Rivals, and Store? Replace the [Name] placeholders.
- ☐ Which storyline direction (A to F), or which mix?
- ☐ Final store name?
- ☐ Paste the class's required GDD sections, so we can map this doc onto them.
- ☐ Do we add the round-winner perk to strengthen inheritance, or keep carts resetting empty?
- ☐ Which Godot version: 4.3 or newer?
- ☐ Due date for the playable build?